

Introduction to Knowledge Representation and Reasoning

Lecture 01 | Introduction to KRR

The knowledge level • representation hypotheses • desirable properties of a KR formalism

Today's roadmap

OVERVIEW

1. Why KRR?

Why intelligent systems need explicit knowledge and reasoning.

2. Knowledge level

Describe a system by what it knows, not only by its code.

3. Representation

How symbols stand for facts, concepts, relations and constraints.

4. Reasoning

How explicit knowledge supports new conclusions and decisions.

5. Good KR formalisms

What makes a representation useful, expressive and computationally workable.

Why does AI need knowledge?

- Intelligent behaviour is not only pattern recognition. An agent must also use what it knows.
- Knowledge lets an AI system interpret observations, connect facts, apply constraints and justify decisions.
- Many useful facts are implicit. Reasoning makes those facts available when needed.
- Knowledge representation gives us a structured, inspectable layer between raw information and action.

Core idea

Knowledge + reasoning → informed action

From the world to intelligent action

MOTIVATION



KRR focuses mainly on the middle: how knowledge is represented and how conclusions are derived from it.

What do we mean by knowledge?

Data

Recorded symbols or measurements.
Example: 85, Machine-7, 14:32

Information

Data interpreted in context.
Example: Machine-7 temperature = 85°C

Knowledge

Meaning plus relationships that support inference.
Example: 85°C exceeds the safe limit → inspection required.

For KRR, the practical question is not “What is knowledge philosophically?” but “How can we represent useful knowledge so a machine can reason with it?”

Knowledge is more than stored facts

Facts: Boiler-1 is operating.

Relations: Boiler-1 supplies steam to Line-A.

Categories: Boiler-1 is a steam-generating asset.

Rules: If pressure exceeds the safe limit, raise an alarm.

Constraints: A machine cannot be both OFF and RUNNING at the same time.

Uncertainty: Sensor evidence suggests a 0.8 probability of bearing failure.

Representation: symbols as surrogates

Real world

Machines, people, objects, events, beliefs, states, constraints and relationships.

REPRESENTS →

Symbolic model

machine(M1)
locatedIn(M1, LineA)
temperature(M1, 85)

A representation is a surrogate for the world. It is useful precisely because it is simpler than the world, but that also means it is always selective.

The representation hypothesis

- An intelligent system can contain explicit symbolic structures that stand for aspects of the world.
- Those structures can be manipulated by computational processes.
- The system's behaviour can therefore be explained partly in terms of what those structures represent.
- Changing the represented knowledge can change behaviour without rewriting every procedure.

Key consequence

Knowledge can be separated from its use.

The knowledge level

Knowledge level

What does the agent know?
What goals does it have?
What follows from that knowledge?

Symbol level

How is knowledge encoded?
Logic? Rules? Graphs? Frames?
Probabilities?

Implementation level

How is it stored and executed?
Data structures, algorithms,
indexes, code.

KRR lets us reason about intelligent behaviour at the knowledge level while still respecting symbol-level expressiveness and implementation cost.

A knowledge-level explanation

KNOWLEDGE LEVEL

System behaviour

The maintenance agent recommends inspecting Pump-4.

Knowledge-level explanation

It knows vibration is high.

It knows high vibration is evidence of mechanical degradation.

It knows Pump-4 is critical.

Therefore it recommends inspection.

This explanation is more meaningful than saying: “function recommend_action() returned INSPECT.”

What is reasoning?

Reasoning uses represented knowledge to derive new conclusions.

Explicit knowledge is what is directly stored.

Implicit knowledge is what follows from what is stored.

Inference procedures are the computational mechanisms that expose useful implicit knowledge.

Simple example

Raining. Raining $\rightarrow$ WetStreet. Therefore: WetStreet.

Explicit vs implicit knowledge

Explicit knowledge

Bird(Tweety)
 $\text{Bird}(x) \rightarrow \text{HasWings}(x)$
 $\text{HasWings}(x) \rightarrow \text{HasBodyPart}(x, \text{Wings})$

Implicit conclusions

$\text{HasWings}(\text{Tweety})$
 $\text{HasBodyPart}(\text{Tweety}, \text{Wings})$

A central KRR problem is deciding which consequences to compute, when to compute them and at what computational cost.

Reasoning is not just deduction

Deduction

Rule + case $\rightarrow$ conclusion

All sensors are devices.

S1 is a sensor.

$\therefore$ S1 is a device.

Abduction

Rule + observation $\rightarrow$ possible explanation

Alarm observed.

Overheat causes alarm.

Overheat may explain it.

Induction

Cases + results $\rightarrow$ general pattern

Observed failures suggest a general relationship.

Default reasoning

Typical rule + no exception $\rightarrow$ tentative conclusion

Birds usually fly.

Different KRR formalisms support different kinds of inference. Choosing a representation also means choosing what kinds of reasoning are natural or possible.

Why formal logic matters

- Formal languages give precise syntax: which expressions are legal?
- Semantics gives meaning: when is an expression true?
- Entailment defines what logically follows from a knowledge base.
- Inference procedures give algorithms for deriving conclusions.
- Soundness and completeness let us evaluate whether a reasoning method is trustworthy.

Desirable properties of a KR formalism

Representational adequacy

Can it express the knowledge the task requires?

Inferential adequacy

Can it support useful new conclusions?

Inferential efficiency

Can reasoning be organized so useful conclusions are found efficiently?

Acquisitional efficiency

Can knowledge be added, changed and maintained conveniently?

Expressiveness comes with a cost

More expressive representation

Can describe richer structures, exceptions and relationships.

But reasoning may become harder or even undecidable.

More restricted representation

May express less.

But inference can be simpler, faster and easier to guarantee.

KRR repeatedly asks us to balance what we want to say against what we can efficiently compute.

One problem, many representations

Propositional logic

$\text{Alarm} \wedge \text{Critical} \rightarrow \text{Evacuate}$

First-order logic

$\text{Critical}(x) \wedge \text{Alarm}(x) \rightarrow \text{Evacuate}(\text{area}(x))$

Rule system

IF alarm AND critical THEN evacuate

Knowledge graph

Pump-4 —locatedIn→ Area-A

Probability

$P(\text{Failure} \mid \text{HighVibration}) = 0.82$

Fuzzy logic

IF temperature is HIGH THEN cooling is STRONG

A running example: intelligent factory assistant

Facts: Machine M7 is on Line 2; current temperature is 92°C.

Rules: If temperature is above 90°C and the machine is running, classify overheating.

Ontology: M7 is a Dryer; Dryer is a ProcessingMachine.

Uncertainty: vibration evidence suggests possible bearing degradation.

Goal: recommend an action and explain why.

KRR question

What should be represented, and how should the agent reason with it?

Where KRR appears in modern AI

APPLICATION

Knowledge graphs

Structured entities, relations and semantic queries.

AI agents

Rules, memory, goals, tools, constraints and planning.

Enterprise AI

Policies, ontologies, workflows and explainable decisions.

Robotics & planning

World models, actions, preconditions and effects.

Hybrid AI

Combining symbolic knowledge with statistical or neural models.

AI-401: where we go from here

- **Logic:** propositional and first-order reasoning
- Horn clauses, Prolog and rule-based systems
- Semantic networks, frames and description logic
- Ontologies, OWL, knowledge graphs and SPARQL
- Logical agents and planning
- Uncertainty, Bayesian networks, fuzzy logic and Markov models
- Advanced and hybrid KRR

Lecture 01 takeaways

- KRR studies how knowledge can be represented so that machines can reason with it.
- A representation is a selective symbolic surrogate for some aspect of the world.
- The knowledge level explains behaviour in terms of what an agent knows and seeks to achieve.
- Reasoning derives useful implicit knowledge from explicit knowledge.
- A good KR formalism balances representational power, inference capability, efficiency and maintainability.

Quick discussion

1. If an AI system stores every fact but cannot infer anything new, does it have useful knowledge?
2. Can the same real-world situation be represented in two different ways? Which one is better?
3. Why might a very expressive representation be a bad engineering choice?
4. Where would you prefer rules over a learned model, and why?

Next lecture

Knowledge, data and code: distinctions and modelling consequences

Reading for Lecture 01

Brachman, R. J. & Levesque, H. J. Knowledge Representation and Reasoning. Morgan Kaufmann, 2004.

Russell, S. & Norvig, P. Artificial Intelligence: A Modern Approach, 3rd ed. Pearson, 2015.

Lecture inspiration: Nebel, B., Lindner, F. & Engesser, T. Principles of Knowledge Representation and Reasoning, Introduction & Organization, 2018.